



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to R.R. AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

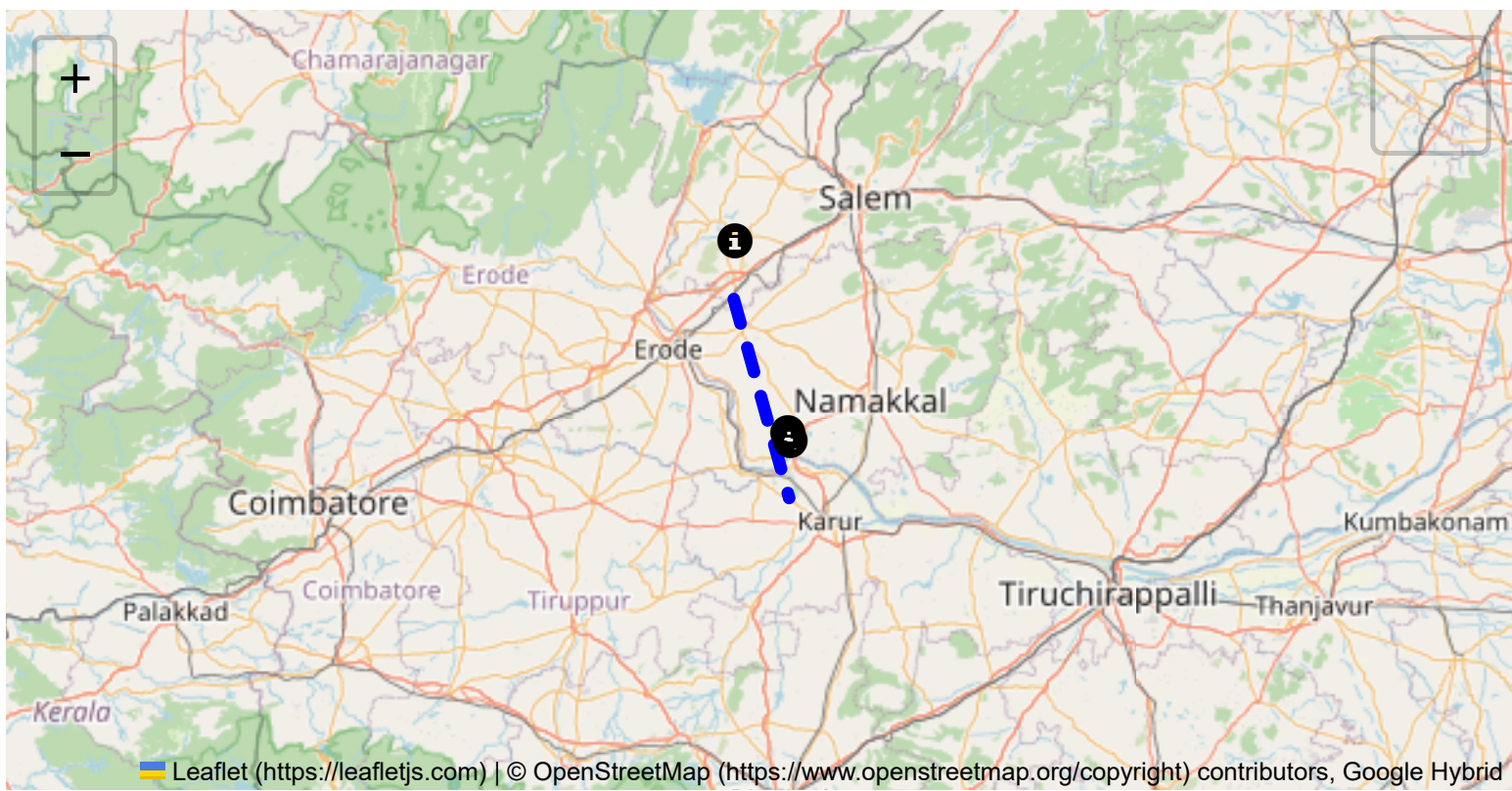
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

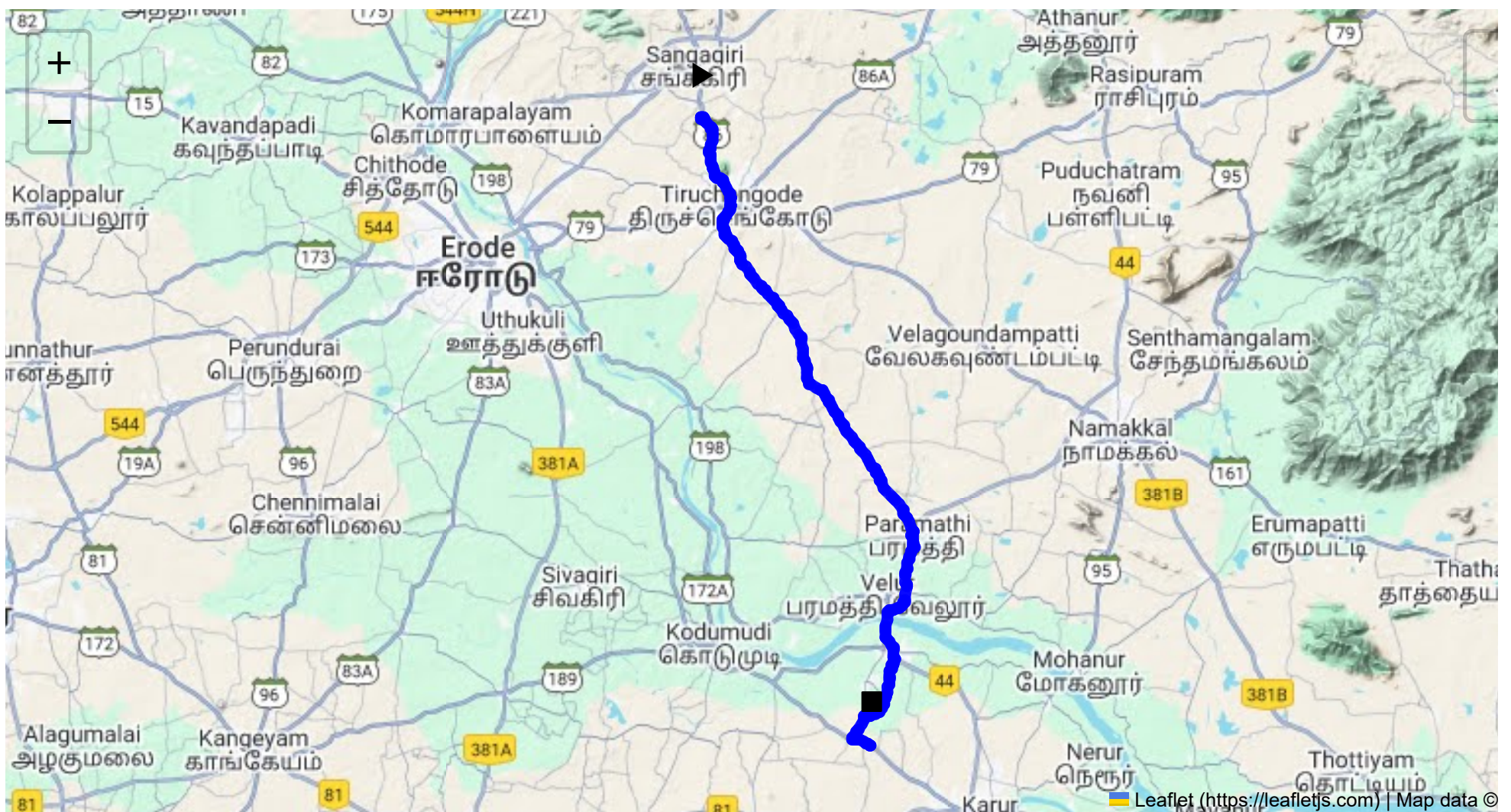
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 57.83 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.008609, 77.991722)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Kagithapuram in Tamil Nadu via TNPL ROAD covers approximately 57.83 kilometers. It passes through rural and semi-urban areas, primarily following state highways and local roads.

Typical Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate with significant rainfall during the northeast monsoon between October and December. The summer months (March to May) can be extremely hot. Rainy seasons might lead to localized flooding and poor road conditions, impacting visibility and vehicle handling. It's essential to monitor weather forecasts before travel.

Traffic Patterns

The route is moderately busy, with peak traffic generally occurring in the early morning (7-9 AM) and late afternoon (5-7 PM) when commuters are traveling to and from work. Congestion might be encountered near town centers and marketplaces, particularly around the Erode-Karur Road.

Road Quality and Infrastructure

Roads on this route are generally in good condition, typical of state highways and major connecting roads in the region. However, some segments, especially rural or less maintained areas, might have potholes or uneven surfaces. Additionally, in some sections, road signs and lane markings may be sparse or faded, requiring attentive driving.

Alternative Routes for Emergencies

In case of obstructions or emergencies, an alternative route would be to redirect via SH189, connecting major roads depending on the situation. Consultation with local traffic updates or GPS navigation services is recommended for real-time adjustments.

Local Regulations for Hazardous Material Transport

Tamil Nadu follows strict regulations concerning the transport of hazardous materials, necessitating permits and adherence to designated routes in populated areas. It is crucial to stay updated with regulatory authorities for any changes affecting these transports.

Historical Incidents

There is limited accessible data on historical incidents specific to this route; however, common issues include accidents related to adverse weather conditions and mechanical failures. Frequent checks on transport equipment and adherence to safety regulations are advised.

Environmental Considerations and Sensitive Areas

The route passes through some agricultural zones that may be sensitive to pollution. Avoid any spillage or accidents that can impact soil and water quality in these areas. Special attention is required when navigating regions close to natural reserves or water bodies.

Communication Coverage

Most areas along the route should have access to mobile communication networks; however, there might be occasional dead zones, particularly in rural or less developed regions. It is advisable to have communication devices fully charged and carry backup options like satellite phones for seamless contact.

Estimated Emergency Response Times

Emergency response in urban or semi-urban areas is generally quicker, ranging from 15-30 minutes. In more remote areas, this can extend to around 45 minutes or longer. Maintaining contact with local emergency services and having a plan for immediate first aid is advisable.

Overall Risk Assessment Summary

This route presents several typical hazards such as traffic congestion, potential poor weather during monsoon seasons, and road condition variabilities. Strict adherence to local regulations when transporting hazardous materials is crucial. Effective planning and constant communication ensure safety and mitigate risks during transit.

Given these insights, truck drivers should prioritize safety checks, remain vigilant about weather conditions, and utilize navigation tools for real-time updates to optimize travel efficiency and safety along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.37868, 77.89498	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16764, 78.01565	30 KM/Hr
7	Turn	High	11.15573, 78.02176	15 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14480, 78.02291	15 KM/Hr
10	Turn	High	11.06822, 78.00969	15 KM/Hr
11	Turn	Medium	11.04255, 78.00324	30 KM/Hr
12	Turn	High	11.04187, 78.00265	15 KM/Hr
13	Turn	High	11.04190, 78.00253	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	Medium	11.03670, 78.00114	30 KM/Hr
15	Turn	Medium	11.03376, 77.99947	30 KM/Hr
16	Turn	Medium	11.03345, 77.99896	30 KM/Hr
17	Turn	High	11.03180, 77.99836	15 KM/Hr
18	Turn	Medium	11.02962, 77.99298	30 KM/Hr
19	Turn	High	11.02933, 77.98831	15 KM/Hr
20	Blind Spot	Blind Spot	11.01327, 77.97999	10 KM/Hr
21	Turn	High	11.00885, 77.99202	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.15573, 78.02176



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.06822, 78.00969



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.04255, 78.00324



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.04187, 78.00265



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04190, 78.00253



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.03670, 78.00114



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.03376, 77.99947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.03345, 77.99896



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.03180, 77.99836



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.02962, 77.99298



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.02933, 77.98831



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.01327, 77.97999



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.00885, 77.99202
